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Return sign forecasts based on conditional risk: Evidence from the UK stock market index

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ABSTRACT

Recent theoretical works have found a link between return sign forecastability and conditional volatility. This paper compares the predictive performance of the conditional country risk and the conditional residual risk in forecasting the direction of change in the return on the UK stock market index. The conditional country risk and the conditional residual risk are estimated using the bivariate BEKK-GARCH technique and the direction of change in the UK stock market index is modelled using the binary logit approach. Both the in-sample and the out-of-sample predictions suggest that, as a predictor, the conditional residual risk is superior to the conditional country risk. Our findings support the residual risk model while contradicting the traditional capital asset pricing model (CAPM). Moreover, our tactical asset allocation simulations show that when the conditional residual risk is used in conjunction with multiple-threshold trading strategies to guide the investment decisions, the actively managed portfolio achieves greater returns than the return on a buy and hold portfolio.

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1. Introduction

In their paper, Christoffersen and Diebold (2006) demonstrate a link between the forecastability of asset return sign and asset return volatility. Their findings suggest that information on conditional volatility can be used to produce direction-of-change forecasts of asset returns. Other authors, including Leung et al. (2000) and Nyberg (2011) for example, provide empirical evidence demonstrating that return sign forecasts are possible and are potentially useful for asset managers who actively allocate their assets between risky and risk-free investments.

Although research on asset return sign predictability is still in its infancy, several research papers in this area are worth mentioning. For example, Christoffersen and Diebold (2006) and Christoffersen et al. (2007) demonstrate a direct link between asset return volatility and asset return sign forecastability. The latter paper also provides empirical evidence suggesting that conditional variance, skewness and kurtosis are all capable of predicting the direction of change in the asset return. Leung et al. (2000) evaluate the efficacy of several multivariate classification techniques relative to a group of level estimation approaches. The classification techniques, which are employed in their work to estimate the probabilities of the direction of change of the return, include linear discriminant analysis, logit, probit and probabilistic neural networks. The authors find that the classification models outperform the level estimation models in terms of predicting the direction of stock market movement and maximising returns from investment trading. Another author who demonstrates that return signs are predictable is Nyberg (2011) who adopts the binary dynamic probit model to investigate the sign predictability of the US stock market. In his paper, the author includes a binary recession indicator, estimated in the first stage using the binary probit model, which is subsequently included in a set of regressors employed in the second stage of the estimation. His findings suggest that the direction of change in the US stock market is predictable with the binary recession indicator being the most useful predictor.

This paper takes the work of Christoffersen et al. (2007), which uses a univariate technique to model conditional volatility, one step further by modelling the conditional asset return volatility using the bivariate BEKK-GARCH technique to capture the impact of time-varying asset return correlation on conditional variance.

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Specifically, we investigate the connection between asset return sign forecastability and conditional asset return volatility using data on the UK stock market index. Our technique also improves upon the work of Nyberg (2011) which calculates the volatility of stock returns as a 1-month rolling window of sum of squared daily stock returns.

The bivariate GARCH approach employed in this article has several attractive features that are worth mentioning. Firstly, it takes into account the impact of correlation between the UK stock market and the global stock market on the conditional return volatility thereby enabling us to capture the 'spill-over' effects between the two markets on the estimated conditional volatility. Secondly, the conditional return volatility and the conditional return correlation are jointly estimated using maximum likelihood estimation which permits asset return volatility to be a function of its own past values as well as its correlation with the other asset return volatility, thereby allowing correlation to change over time. The implication of time-varying correlation cannot be understated as a large body of research has shown that correlation is indeed unstable and changes over time (see Fabozzi and Francis (1978), Bos and Newbold (1984), Faff et al. (1992) and Brooks et al. (2002), for example). Thirdly, as the approach allows for time-varying correlation, it is possible for us to examine whether conditional country risk, which is a measure of time-varying co-movement between the return on a country's stock market and the return on the world stock market, can be used to forecast the direction of change in the asset return. In order words, we examine whether the main result of the Capital Asset Pricing Model (CAPM), which suggests that a return on the asset should reflect only the undiversifiable risk component (i.e., the systematic risk), is valid when the direction of change in the asset return instead of the level is modelled.

Our research method involves two stages. In the first stage, the bivariate BEKK-GARCH is fitted to the data to estimate the conditional variance–covariance matrices. The conditional country risk (also called the conditional beta or the time-varying beta) and the conditional volatility (also called the conditional residual risk or the conditional idiosyncratic risk) are computed using the information obtained from the conditional variance–covariance matrices. The two measures of risk are subsequently used as explanatory variables in the second-stage estimation. In the second stage, we employ the binary logit model to predict the direction of change in the UK stock market return. In our model, only two possible states for the change in the return are possible: 'upwards' and 'downwards'. A set of diagnostic tests are performed to select a benchmark model whose predictive performance serves as a yardstick to which the predictive performance—both in-sample and out-of-sample—of the other models may be compared. In addition, using the out-of-sample return sign forecasts, we simulate investment returns generated by different trading strategies: the passive 'buy and hold' strategy and the active asset allocation strategies guided by the benchmark model, the conditional country risk model and the conditional residual risk model. This type of return simulation is potentially beneficial to asset managers whose job is to allocate their managed assets between risky and risk-free rate investments. Incorporated into our simulations are transaction costs and threshold trading rules employed by Leung et al. (2000). These rules are designed to better utilise the probability forecasts, generated by the binary logit estimations, by taking into account 'degrees of confidence' in the forecasting models' predictive strength.

Our results suggest that both the conditional country risk and the conditional volatility are capable of predicting directional changes in returns on the UK stock market. Specifically, the conditional residual risk possesses statistically significant predictive power, generating superior return sign forecasts both in-sample and out-of-sample. Diagnostic tests based on log-likelihood functions and outcome-based predictive performance evaluation both confirm the usefulness of the conditional residual risk in predicting the direction of change in the return on the UK stock market index. Our findings add to the growing literature on the relevance of residual risk to asset pricing. Furthermore, our investment return simulation results suggest that the return on a portfolio managed actively using the probability forecasts obtained from the conditional residual risk model exceeds the return on a passive 'buy and hold' portfolio provided that multiple threshold trading rules are employed.

The article proceeds as follows. We briefly present the theoretical relationship between asset return volatility and return sign forecastability as well as relevant literature on usefulness of residual risk in Section 2. We then outline our methodology—the BEKK-GARCH technique and the binary logit model—in Section 3. While Section 4 explains the data, we present both the BEKK-GARCH and the binary logit estimation results in Section 5. The in-sample and out-of-sample results, as well as forecast evaluations, are discussed in Section 6. Section 7 deals with trading simulations. Finally, Section 8 concludes the paper.

2. Volatility dynamics and direction-of-change forecasts

The ability of volatility to predict asset return signs is derived from the interaction between a nonzero mean return and nonconstant volatility and is independent of the shape of the return distribution. However, when the distribution of asset returns is symmetric, either a zero mean or constant volatility would render the sign unforecastable as pointed out by Christoffersen and Diebold (2006). For the same reason, when volatility is large relative to the mean return, the return sign also becomes nearly unpredictable: in a highly volatile market, forecasting where the market will go next is difficult. Christoffersen and Diebold (2006) also indicate that sign forecastability is often disguised by a practice of removing unconditional means and working with zero-mean series in traditional financial econometrics.

In this section, we explain in more detail the concepts of conditional mean dependence, sign dependence and volatility dependence. We then review and discuss in the following subsection relevant literature that focuses on investigating the extent to which estimated conditional volatility can be used to explain asset returns.

2.1. Conditional mean dependence, sign dependence and volatility dependence

According to Christoffersen and Diebold (2006), approximate conditional mean independence in asset returns, asset return sign dependence and asset return volatility dependence are interrelated. To explore this idea further, let X_t be the information set available to the investor at time t. Conditional mean return independence implies that $E(R_{t+1}|X_t)$ does not vary with X_t. Furthermore, when R_{t+1} displays sign forecastability (sign dependence), the expected return sign indicator series $E(I(R_{t+1} > 0)|X_t)$ varies with X_t; and when R_{t+1} displays conditional variance dependence (also referred to as conditional variance dynamics), $\sigma^2_{t+1|t} = \mathrm{Var}(R_{t+1}|X_t)$ varies with X_t.

The link between conditional return volatility and return sign forecastability can be demonstrated as follows. Consider the return on a risky asset at time $t + 1$ conditional upon the information set X_t:

$$R_{t+1}|X_t \sim N\left(\mu, \sigma^2_{t+1|t}\right) \tag{1}$$

where $\mu = E(R_t) = E(R) > 0$ and $\sigma^2_{t+1|t} = \mathrm{Var}(R_{t+1}|X_t)$. While the return is assumed to be conditional mean independent (i.e., not

time-varying), it displays conditional variance dependence. [1] Define the probability of a positive return: $\Pr(R_{t+1} > 0) = 1 - \Pr(R_{t+1} < 0)$. It can be shown that:

$$\Pr(R_{t+1} > 0) = 1 - \Pr\left(\frac{R_{t+1} - \mu}{\sigma_{t+1|t}} < \frac{-\mu}{\sigma_{t+1|t}}\right) = 1 - \Phi\left(\frac{-\mu}{\sigma_{t+1|t}}\right) = \Phi\left(\frac{\mu}{\sigma_{t+1|t}}\right)$$

where $\Phi(\cdot)$ is the $N(0,1)$ cumulative density function. Given that $\mu > 0$ by our assumption, the fact that $\Pr(R_{t+1} > 0) > 0.50$ implies forecastability of the direction of change in the asset return. Notice that if we assume that the return distribution is asymmetric, the sign can still be predictable even if $\mu = 0$.

2.2. Conditional residual risk and conditional systematic risk

The Capital Asset Pricing Model (CAPM), independently developed by Sharpe (1964), Lintner (1965) and Mossin (1966), states that an investor is not rewarded by taking on additional asset-specific risk (also called residual risk or idiosyncratic risk) because the risk is diversifiable by increasing the holding of assets with low or negative correlations in the portfolio. In theory, a well-diversified portfolio with no residual risk is still exposed to systematic risk which measures the extent to which returns on the portfolio vary with returns on the theoretical market portfolio. A portfolio with higher beta is said to have larger systematic risk and should generate higher returns than a portfolio constructed to have a smaller beta. Under an international diversification framework, the value of beta measures the extent to which returns on a country stock market index vary with returns on the world stock market index. This measure, called country risk, has been shown to fluctuate over time. Empirical evidence suggesting that country risk is time-varying can be found in Fabozzi and Francis (1978), Sunder (1980), Bos and Newbold (1984), Harvey (1989), Bodurtha and Mark (1991), Ferson (1989), Faff et al. (1992), Giannopoulos (1995), Gonzalez-Rivera (1996) and Brooks et al. (2002), among others.

As opposed to the static CAPM, which assumes that country risk is constant, the conditional CAPM allows the risk to change over time:

$$E(r_{i,t}|X_{t-1}) = \beta_{i,t}[E(r_{m,t}|X_{t-1})] \tag{2}$$

where $r_{i,t}$ and $r_{m,t}$ are the excess return on a country's stock market and the excess return on the world stock market at time t, respectively; $E(\cdot|X_{t-1})$ is the expectation operator conditional on the information set X available to the investor at time $t-1$. From the investor's perspective, $\beta_{i,t}$ measures the contribution of the country's stock market risk to the total risk of the world stock market and can be estimated using the following equation:

$$\beta_{i,t} = \frac{\sigma_{im,t}}{\sigma^2_{m,t}} \tag{3}$$

where $\sigma_{im,t}$ is the covariance between the excess return on the country's stock market and the excess return on the world market and $\sigma^2_{m,t}$ is the variance of the excess return on the world stock market.

Research on the impact of time-varying risk—both systematic and asset-specific risk—on asset returns is abundant, although the findings are somewhat inconclusive and often contradictory. While a large body of research points to a statistically significant relation between asset returns and conditional systematic risk, several papers conclude that idiosyncratic risk is more relevant to asset pricing, thus rejecting the role of systematic risk and, by implication, the CAPM. [2]

As far as econometric approaches for estimating risk are concerned, the Schwert and Seguin (1990) method, the Kalman filter approach, stochastic volatility (SV) models (Johansson, 2009) and GARCH techniques are among the most popular choices. In terms of popularity, however, GARCH approaches have received by far the most attention. In addition, the performance of GARCH methods are also shown to be superior to the other methods when evaluated using some standard error metrics (see Brooks et al., 2002). We review several research articles that explore the connection between systematic risk, residual risk and asset returns below.

Bollerslev et al. (1988) pioneer the use of a multivariate GARCH approach to test the CAPM where expected returns are assumed to be a function of a time-varying covariance matrix of returns on three assets: bonds, bills and stocks. Their results, which suggest that the conditional covariances are not stable, are the first to show that systematic risk is not constant but is time-varying. Moreover, they find that the conditional covariances can be used to explain variations in the returns.

Ng (1991) subsequently utilises a similar multivariate GARCH framework, adopted by Bollerslev et al. (1988), with a constant correlation (CC) assumption to test the CAPM. She uses data on monthly returns on all common stocks traded in the New York Stock Exchange (NYSE) over the period January 1926 to December 1987. The results suggest that whether or not the CAPM is rejected depends on how portfolios of assets are constructed. The CAPM is not rejected when portfolios are formed by ranking betas, but constructing portfolios according to size leads to a rejection of the CAPM.

Glosten et al. (1993) investigate the relation between risk and return by using a GARCH-M approach to model time-varying volatility. Their results point to a negative relationship between conditional expected monthly returns and the conditional monthly variance. The model is revised to allow positive and negative unexpected returns to affect conditional variances in a different manner. They find the negative relation to be stronger and statistically significant when the conditional variance is allowed to have deterministic monthly seasonals and to depend on the nominal risk-free interest rate.

Similar to Glosten et al. (1993), Hansson and Hordahl (1998) also use a bivariate GARCH-M technique to test the conditional CAPM against six alternative specifications, which include the zero-beta version of CAPM, a conditional residual risk model and models which nest the international CAPM and the consumption CAPM. They estimate the models using data on returns on the Swedish stock market for the period between 1977 and 1990. Their findings support the conditional Sharpe–Lintner–Mossin CAPM and reject the other model specifications.

A statistically significant relation between the average stock variance and the return on the market portfolio is found by Goyal and Santa-Clara (2003). In their paper, they argue that background risk is added to investors' traded portfolios from their holding of non-traded assets. An increase in the risk of the nontraded assets causes investors to require an increase in expected return in order to be persuaded to hold the market portfolio of traded stocks. This leads to a tradeoff between market return and average stock price.

Guo and Savickas (2003) compute new measures of realised value-weighted idiosyncratic volatility (IV) and realised stock

[1] Christoffersen and Diebold (2006) point out that conditional mean independence is not a *pre-requisite* for the relation between sign dependence and volatility dependence. The assumption of conditional mean independence is invoked to make the point as clearly as possible.

[2] Another strand of literature also employs the international CAPM to investigate the degree of capital market integration by looking at the impact of the conditional variance on asset returns. For more information on this class of literature, see Bekaert and Harvey (1995), Carrieri et al. (2007) and Guesmi and Nguyen (2011), for instance.

market volatility (MV) using daily data from July 1962 to December 2002. The joint effects of IV and MV on excess stock market returns are examined and found to possess strong predictive power. The relationship between risk and return is found to be positive, as suggested by the CAPM. The link between idiosyncratic risk and future stock market returns, however, is found to be negative.

Ghysels et al. (2004) introduce a new risk measure that models the monthly variance using past daily squared returns. This approach is called Mixed Data Sampling (MIDAS). Using both monthly and daily market return data between 1964 and 2000, they find a positive and significant relationship between risk and return. They also point out that, when compared to negative shocks, positive shocks have a bigger overall impact on the conditional mean return and are slower to be incorporated into the conditional variance. Furthermore, positive shocks are found to be much more persistent and account for the persistent nature of the conditional variance process.

Finally, Fu (2009) employs EGARCH to estimate the conditional idiosyncratic volatilities and finds that the conditional idiosyncratic risk possesses predictive power and is capable of explaining movements in the expected return. The positive relationship between risk and return is found to be both economically and statistically significant and is robust to different testing methods.

3. Methodology

Our methodology involves two stages. In the first stage, we estimate the conditional country risk and the conditional residual risk for the UK stock market using the bivariate BEKK-GARCH approach. The BEKK-GARCH technique is appealing due to its ability to jointly model volatility clustering and correlation clustering—a phenomenon often observed in the financial markets. Owing to the difficulty of estimating parameters in a system of equations, the BEKK-GARCH approach greatly simplifies estimation as it parameterises the GARCH equations to ensure positive definiteness of the variance–covariance matrix. The restrictions guarantee convergence when the model is estimated by maximum likelihood.

In the second stage, we employ the binary logit technique to model the direction of change in the UK stock market return. The conditional country risk and the conditional return volatility, estimated using the BEKK-GARCH approach in the first stage, enter the binary logit estimations as explanatory variables. We control for the effects of other factors, often thought to be useful in predicting directional changes in stock returns, by including short and long interest rates, term spreads, the output gap and balance-sheet variables in the explanatory variable sets.

3.1. The BEKK-GARCH model

The basic BEKK-GARCH specification, described in Engle and Kroner (1995), is defined as follows:

$$\mathbf{r}_t = \boldsymbol{\varepsilon}_t \tag{4}$$

where

$$\boldsymbol{\varepsilon}_t = \mathbf{H}_t^{1/2}\mathbf{z}_t \tag{5}$$

where

$$\mathbf{H}_t = \mathbf{C}'\mathbf{C} + \sum_{i=1}^{q}\mathbf{A}_i'\boldsymbol{\varepsilon}_{t-i}\boldsymbol{\varepsilon}_{t-i}'\mathbf{A}_i + \sum_{j=1}^{p}\mathbf{B}_j'\mathbf{H}_{t-j}\mathbf{B}_j \tag{6}$$

The matrices **A**, **B** and **C** are $n \times n$ matrices and $\mathbf{C}'\mathbf{C}$ is assumed to be either upper or lower triangular and where n is the number of assets to be modelled. For example, in a bivariate case where $n = 2$, the BEKK-GARCH(1,1) can be explicitly written as follows:

$$\begin{bmatrix} \sigma_{i,t}^2 & \sigma_{im,t} \\ \sigma_{mi,t} & \sigma_{m,t}^2 \end{bmatrix} = \begin{bmatrix} c_{11} & 0 \\ c_{21} & c_{22} \end{bmatrix} + \begin{bmatrix} a_{11} & a_{21} \\ a_{12} & a_{22} \end{bmatrix} \begin{bmatrix} \varepsilon_{i,t-1}^2 & \varepsilon_{i,t-1}\varepsilon_{m,t-1} \\ \varepsilon_{m,t-1}\varepsilon_{i,t-1} & \varepsilon_{m,t-1}^2 \end{bmatrix} \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} + \begin{bmatrix} b_{11} & b_{21} \\ b_{12} & b_{22} \end{bmatrix} \begin{bmatrix} \sigma_{i,t-1}^2 & \sigma_{im,t-1} \\ \sigma_{mi,t-1} & \sigma_{m,t-1}^2 \end{bmatrix} \begin{bmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \end{bmatrix} \tag{7}$$

When Eq. (7) is employed to model the time-varying risk in the UK stock market, we use the subscripts i and m to denote the UK stock market variables and the world stock market variables respectively. Notice that the BEKK-GARCH model specifies the conditional variance of one variable to be a function of lagged values of the other variables. The model can be thought of as a restricted version of the VEC-GARCH model, originally proposed by Bollerslev et al. (1988), where the conditional variance–covariance matrices are constructed to be positive definite. The specification guarantees convergence when the model is estimated by maximum likelihood. The multivariate GARCH log-likelihood function can be found in Appendix A.1.

3.2. The binary logit model

Since directional changes in returns are either 'downwards' or 'upwards', we construct a random variable q_t to take the value of 0 and 1 according to the following criteria: $q_t = 0 \Leftrightarrow r_{i,t} \leqslant 0$ and $q_t = 1 \Leftrightarrow r_{i,t} > 0$ where $r_{i,t}$ is the excess return on the UK stock market index at time t. Let X_t be a $k \times 1$ vector of explanatory variables available to the investor at time t, with the first element of X_t set equal to one. The probability of $q_t = 1$ conditional on X_t is defined by the logit cumulative density function:

$$\Pr(q_t = 1|X_t) = \frac{e^{X_t'\beta}}{1 + e^{X_t'\beta}} \tag{8}$$

and $\Pr(q_t = 0|X_t) = 1 - \Pr(q_t = 1|X_t)$ where β is the unknown $k \times 1$ parameters to be estimated by maximum likelihood. The binary logit log-likelihood function can be found in Appendix A.2.

The predicted probabilities $\widehat{P}_0$ and $\widehat{P}_1$ can be calculated once we obtain the estimated parameters $\hat{\beta}$. Our prediction $\hat{q}_t$ is given by the following decision rule:

$$\hat{q}_t = m \Longleftrightarrow \widehat{P}_m = \max(\widehat{P}_0, \widehat{P}_1) \tag{9}$$

Notice that Eq. (9) uses a single threshold of 0.50 to decide if the direction of change is 'upwards' or 'downwards'.

4. Data

We use monthly data spanning a period of 16 years. Our sample starts in January 1996 and ends in October 2011, giving a total of 190 observations. The returns on the UK stock market index and the world stock market index at time t, denoted by $R_{i,t}$ and $R_{m,t}$, are calculated as month-to-month percentage changes on the Morgan Stanley Capital International (MSCI) Country Index and the MSCI World Index, respectively.[3] Both indices are denominated in

[3] The MSCI World Index is a free float-adjusted market capitalization weighted index that is designed to measure the equity market performance of developed markets. The MSCI World Index consists of the following 24 developed market country indices: Australia, Austria, Belgium, Canada, Denmark, Finland, France, Germany, Greece, Hong Kong, Ireland, Israel, Italy, Japan, Netherlands, New Zealand, Norway, Portugal, Singapore, Spain, Sweden, Switzerland, the United Kingdom, and the United States.

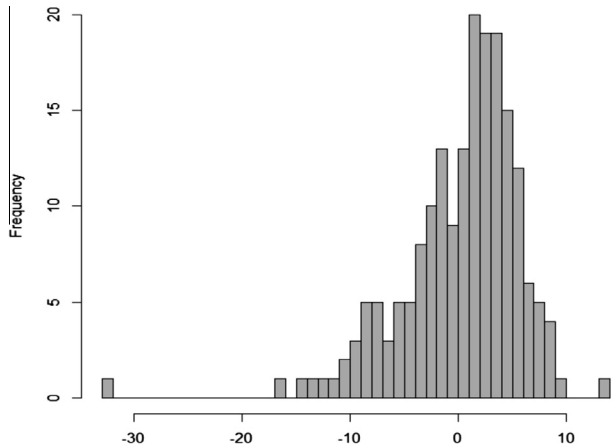

Fig. 1a. Histogram of the returns on the UK stock market index.

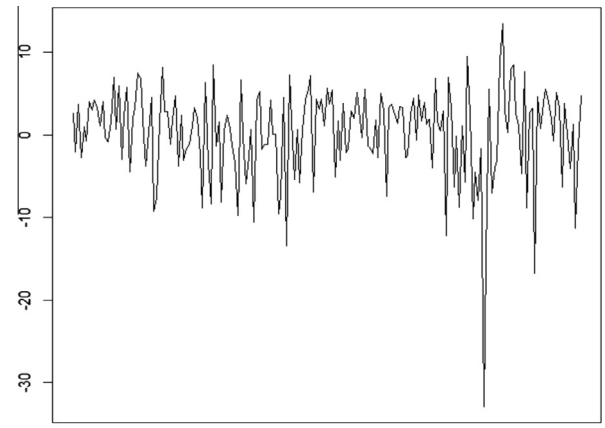

Fig. 1b. The returns on the UK stock market index.

US dollars. We use the 3-month US Treasury bill rate as a proxy for the risk-free rate, which we denote by $R_{f,t}$. The time series of the monthly risk-free rate is calculated from the annual risk-free rate using geometric compounding.[4] The excess return on the UK stock market index and the excess return on the world stock market index at time t, denoted by $r_{i,t}$ and $r_{m,t}$, are calculated as follows: $r_{i,t} = R_{i,t} - R_{f,t}$ and $r_{m,t} = R_{m,t} - R_{f,t}$. The histogram and the time-series plot of the returns on the UK stock market index are shown in Figs. 1a and 1b, respectively.

We use 3-month and 10-year UK government bond yields as proxies for the short-term and long-term interest rates, denoted by r_t^S and r_t^L. The term spread at time t, denoted by SP_t, is calculated as the difference between the short-term interest rate and the long-term interest rate: $SP_t = r_t^L - r_t^S$. The performance of the real sector of the economy is captured by the output gap, denoted by $\tilde{y}_t$. To calculate the output gap, we detrend the index of industrial production using the Hodrick–Prescott filter. The output gap at time t is calculated as the percentage deviation of the actual output from the trend: $\tilde{y}_t = 100 \times \left[(y_t - y_t^*)/y_t^*\right]$, where y_t and y_t^* are the actual level of output and the trend output at time t. We include two balance sheet variables, the dividend yield and the earnings yield—often documented in the finance literature for its ability to help explain variations in asset returns—in our models to help predict the direction of change in the return on the UK stock market. They are denoted by DY_t and EY_t respectively.

Our choice of explanatory variables is motivated by previous studies on asset pricing and asset return sign forecasts. For instance, the first difference of the short-term rate is found to possess explanatory power by Leung et al. (2000), while the short rate is proved to be a useful predictor by Ang and Bekaert (2007). Moreover, Nyberg (2011) finds the first difference of the long rate and the first difference of the short rate to have some explanatory power. The term spread, which is useful in identifying recession periods (see Bernard and Gerlach, 1998; Estrella and Mishkin, 1998), also helps predicting stock returns as returns are likely to be negative during recessions. The dividend yield is commonly used as an explanatory variable by Bekaert and Harvey (1995), Pesaran and Timmerman (1995), Ang and Bekaert (2007) and Nyberg (2011), while the earnings yield is shown to be a powerful predictor of the return on stock by Campbell and Shiller (1988), for example.

All the variables, except for the 3-month US Treasury bill rate, were obtained from Datastream. The 3-month US Treasury bill rate series was downloaded from the website of the Board of Governors of the Federal Reserve System.[5]

5. Estimation and results

We begin our analysis by testing for the presence of ARCH effects in both the raw returns and the estimated residuals using the Lagrange multiplier (LM) test first derived by Engle (1982). The chi-squared test statistics calculated on the excess returns on the UK stock market index series and the excess returns on the global market index series equal 3.26 and 5.56 with the p-values of 0.07 and 0.02 respectively. At conventional levels of statistical significance, both the null hypothesis of no ARCH in the returns on the UK stock market index series and the null hypothesis of no ARCH in the returns on the world market index series are rejected. As for the estimated residuals, the test statistics for the estimated residuals of the UK return series and the world return series equal 2.50 and 3.15 with p-values of 0.11 and 0.08 respectively. While the null hypothesis of no ARCH in the residuals of the global return series is rejected at the 10% significance level, the null hypothesis of no ARCH in the estimated residuals of the UK market index is not rejected inspite of the fact that the p-value of 0.11 is very close to the 0.10 rejection threshold for the 10% significance level. We now proceed to the next stage of our analysis on the grounds that the LM tests indicate to some extent the presence of ARCH in the return series.[6]

In the next step, we fit the BEKK-GARCH model to the data on the UK stock market and the global stock market to estimate the conditional variance–covariance matrices. We estimate a total of four BEKK-GARCH models to identify the best-fitting specification: BEKK-GARCH(1,1), BEKK-GARCH(1,2), BEKK-GARCH(2,1) and BEKK-GARCH(2,2). We select the best-fitting model using the AIC statistic.

Table 1 shows the BEKK-GARCH estimation results along with the AIC statistics. The BEKK-GARCH(2,1) appears to fit the data

[4] The monthly risk-free rate at time t is calculated as follows:

$$R_{f,t} = 100 \times \left[\left(1 + \frac{R_t^*}{100}\right)^{\frac{1}{12}} - 1\right]$$

where $R_{f,t}$ and R_t^* denote the monthly risk-free rate and the 3-month Treasury bill rate at time t respectively.

[5] http://www.federalreserve.gov/.

[6] The chi-squared test statistic is calculated as follows:

$$LM = TR^2$$

where R^2 is the goodness-of-fit measure from the regression of squared returns (residuals) on a constant and p lags of squared returns (residuals). In our tests, we set $p = 1$.

Table 1
The BEKK-GARCH estimation results.

Coefficients	BEKK-GARCH (1,1)	BEKK-GARCH (1,2)	BEKK-GARCH (2,1)	BEKK-GARCH (2,2)
c_1	0.8867	0.9373**	0.5618	0.7778
	(0.5886)	(0.3061)	(0.3671)	(0.5749)
c_2	0.7428	1.4020**	1.2643**	1.3132
	(0.7618)	(0.3307)	(0.3439)	(0.5095)
c_3	0.9559**	0.0000	−0.0002	0.0016
	(0.2749)	(0.9188)	(1.3244)	(0.8961)
a_1	0.4475*	−0.0319	0.4939**	0.7003
	(0.2291)	(0.1908)	(0.2031)	(0.1945)
a_2	−0.1011	0.3744*	−0.0649	0.2360
	(0.2481)	(0.2026)	(0.1939)	(0.2081)
a_3	−0.1594	0.1651	−0.0722	−0.3019
	(0.1948)	(0.1603)	(0.1596)	(0.1596)
a_4	0.3421*	0.2544	0.3861**	0.0767
	(0.2048)	(0.1741)	(0.1505)	(0.1705)
a_5		−0.0009		−0.1099
		(0.2101)		(0.2122)
a_6		−0.0792		−0.0521
		(0.2286)		(0.2098)
a_7		0.2450		0.4292
		(0.1606)		(0.1793)
a_8		0.2544		0.4719
		(0.1991)		(0.1696)
b_1	0.7558**	0.3061	−0.1733	−0.3021
	(0.1613)	(0.4363)	(0.4191)	(0.4075)
b_2	−0.0183	1.2636**	0.3411	0.4378
	(0.1499)	(0.4047)	(0.3887)	(0.2883)
b_3	0.1856	0.6070	0.9525**	0.0993
	(0.1617)	(0.3859)	(0.3305)	(0.4731)
b_4	0.9598**	−0.2715	0.6247*	−0.6314
	(0.1379)	(0.4295)	(0.3556)	(0.3152)
b_5			−0.3734	−0.0089
			(0.3827)	(0.4239)
b_6			0.3071	0.4917
			(0.3653)	(0.4132)
b_7			0.4388	0.7557
			(0.2948)	(0.3538)
b_8			−0.2032	0.3517
			(0.3665)	(0.3843)
AIC	1002.262	1001.496	1000.420	1000.925

Note: Standard errors are shown in parentheses. A total of 189 observations are used in the BEKK-GARCH estimations. The sample period starts from 02/1996 to 10/2011.
The symbol * indicates statistical significance at the 10% significance level.
The symbol ** indicates statistical significance at the 5% significance level.

best. Using the estimated coefficients, we recursively compute the conditional residual risk and the conditional country risk, which we denote by σ_t^2 and β_t henceforth. The latter is computed according to Eq. (3) using the fitted values of $\sigma_{m,t}^2$ and $\sigma_{im,t}$ obtained from the estimated conditional variance–covariance matrices. The mean value of the conditional country risk in our sample equals 0.98. It is

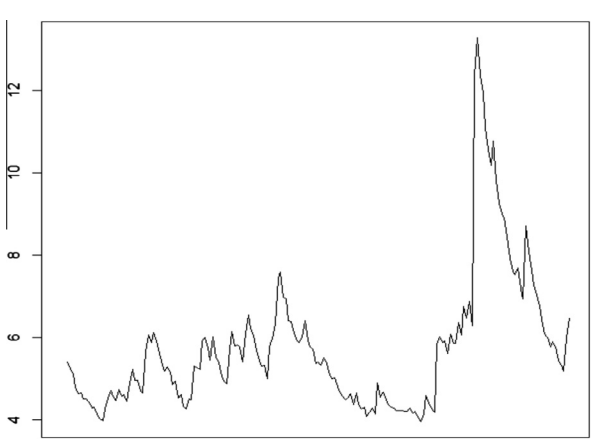

Fig. 1c. Conditional volatility.

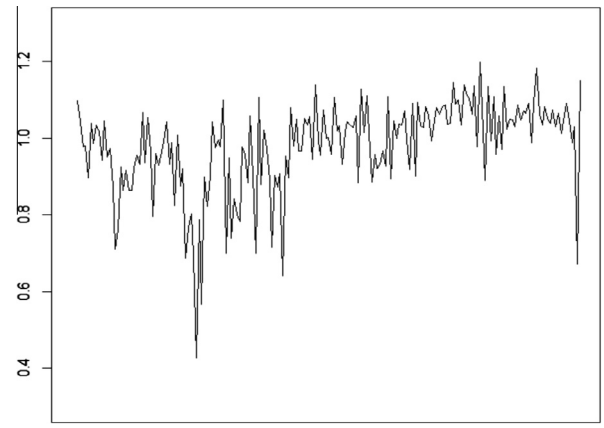

Fig. 1d. Conditional beta.

expected that the value should be close to 1 as the UK stock market index represents a relatively large proportion of the world stock market index. The time-series plots of the conditional volatility and the conditional beta are shown in Figs. 1c and 1d, respectively.

We employ the binary logit model in the second stage to evaluate the direction of change in the return on the UK stock market. As explained in Section 3.2, the dependent variable q_t in the binary logit model captures the return sign, taking the values of 0 when the return sign is negative and 1 otherwise. A total of six specifications are estimated; the standard errors for the binary logit coefficients are bootstrapped to correct for bias since both of the risk variables—the conditional residual risk and the conditional country risk—are fitted values calculated in the first-stage estimation. Following Pesaran and Timmerman (1995), we include only the information most recently available to the investor in our models as well as changes in both the short rates and the long rates in the estimations. We discuss each specification below.

Model (1), denoted $X_{1,t}$, comprises the lagged value of the term spread, the lagged first difference of the short-term interest rate and the lagged first difference of the long-term interest rate:

$$X_{1,t} = \left(1, SP_{t-1}, \Delta r_{t-1}^S, \Delta r_{t-1}^L\right)'.$$

To control for the impact of real activity on the UK stock market, Model (2), denoted $X_{2,t}$, adds to Model (1) the lagged output gap:

$$X_{2,t} = \left(1, SP_{t-1}, \Delta r_{t-1}^S, \Delta r_{t-1}^L, \tilde{y}_{t-1}\right)'.$$

Model (3), denoted $X_{3,t}$, adds the lagged first difference of the dividend yield to Model (2):

$$X_{3,t} = \left(1, SP_{t-1}, \Delta r_{t-1}^S, \Delta r_{t-1}^L, \tilde{y}_{t-1}, \Delta DV_{t-1}\right)'$$

while Model (4), denoted $X_{4,t}$, adds the lagged first difference of the earnings yield to Model (3):

$$X_{4,t} = \left(1, SP_{t-1}, \Delta r_{t-1}^S, \Delta r_{t-1}^L, \tilde{y}_{t-1}, \Delta DV_{t-1}, \Delta EY_{t-1}\right)'.$$

As preliminary analyses, we estimate Models (1)–(4) using the binary logit specification. The estimation results are shown in columns (1)–(4) in Table 2. We test for the joint significance of the slope parameters in β by employing the statistic

$$2(\ln L_{\mathrm{UN}} - \ln L_{\mathrm{RS}}) \sim \chi_c^2 \tag{10}$$

where $\ln L_{\mathrm{UN}}$ is the value of log-likelihood function of the unrestricted (or full) model, $\ln L_{\mathrm{RS}}$ is the value of log-likelihood function of the restricted version of the model and c is the number of restrictions imposed on the restricted model respectively.[7] We measure the

[7] The restricted version of the model contains only the intercept term when we test for the joint significance of all the slope parameters.

Table 2
The binary logit estimation results.

Variables	Model					
	(1)	(2)	(3)	(4)	(5)	(6)
constant	−0.010	0.041	0.063	0.066	0.059	0.089
	(0.165)	(0.167)	(0.171)	(0.184)	(0.187)	(0.178)
SP_{t-1}	0.022	−0.058	−0.038	−0.033	−0.040	0.237
	(0.108)	(0.122)	(0.141)	(0.137)	(0.136)	(0.134)
Δr^S_{t-1}	−0.414	−0.176	−0.215	−0.274	−0.157	−0.126
	(0.843)	(0.872)	(0.972)	(1.042)	(0.963)	(0.898)
Δr^L_{t-1}	−0.563	−0.655	−1.098	−1.154	−1.175	−0.952
	(0.745)	(0.749)	(0.815)	(0.861)	(0.853)	(0.812)
$\tilde{y}_{t-1}$		−0.2008*	−0.364**	−0.345**	−0.3614**	−0.376**
		(0.114)	(0.147)	(0.138)	(0.142)	(0.133)
ΔDV_{t-1}			3.977**	5.280**	3 997**	3.822**
			(1.26)	(1.645)	(1.196)	(1.055)
ΔEY_{t-1}				−0.677		
				(0.606)		
$\Delta\beta_t$					1.251	
					(1.415)	
$\Delta\sigma_t$						1.237**
						(0.544)
$\log L_{RS}$	−128.208	−128.208	−128.208	−128.208	−128.208	−128.208
$\log L_{UN}$	−127.465	−125.743	−114.872	−114.016	−114.346	−111.905
Chi-squared	1.486	4.930	26.6719**	28.38426**	27.7232**	32.6067**
Prob.	0.686	0.295	0.000	0.000	0.000	0.000
McFadden R^2	0.006	0.019	0.104	0.111	0.108	0.127
Estrella R^2	0.008	0.027	0.141	0.150	0.147	0.172
AIC	1.421	1.413	1.307	1.308	1.312	1.285

Note: Standard errors are shown in parentheses. A total of 185 observations are used in the logit estimations. The sample period starts from 06/1996 to 10/2011.
The symbol * indicates statistical significance at the 10% significance level.
The symbol ** indicates statistical significance at the 5% significance level.

goodness-of-fit of the models using two pseudo-R^2 measures, suggested by McFadden (1974) and Estrella (1998). Similar to the conventional R^2 statistic in linear regression, both measures are bounded by 0 and 1. The pseudo-R^2 measures have the following functional forms:

$$R^2_{\text{McFadden}} = 1 - \frac{\ln L_{\text{UN}}}{\ln L_{\text{RS}}} \tag{11}$$

$$R^2_{\text{Estrella}} = 1 - \left(\frac{\ln L_{\text{UN}}}{\ln L_{\text{RS}}}\right)^{(-2\ln L_{\text{RS}}/T)} \tag{12}$$

While the pseudo-R^2 proposed by McFadden (1974) is a better known measure and has been around for a long time, the measure suggested by Estrella (1998) is more closely in line with the conventional R^2 used in the linear model due to its marginal relationship with the average likelihood ratio statistic.

The result in column (1) indicates that none of the coefficients are statistically significant. The chi-squared test for the joint significance of the slope parameters fails to reject the null hypothesis that all the slope parameters equal to zero, suggesting that Model (1) possesses no predictive power. Both the McFadden R^2 and the Estrella R^2 statistics are very low, indicating a poor fit. In column (2), only the coefficient on the lagged output gap is statistically significant but only at the 10% significance level. However, the result for Model (3), which adds the lagged first difference of the dividend yield to Model (2), rejects the null hypothesis of no predictability, suggesting that the variables in Model (3) can jointly explain the direction of change in the return on the UK stock market. The coefficients on both the lagged output gap and the lagged first difference of the dividend yield in column (3) are statistically significant at the 5% significance level. The pseudo-R^2 and the AIC statistics for Model (3) indicate a significant improvement in the goodness of fit over Models (1) and (2). As for Model (4), the estimation result indicates no significant improvement from Model (3) when the lagged first difference of earning yield is included in the regressor set. Interestingly, although we find that the dividend yield possesses predictive power, the result of which is supported by many previous studies, the interest rate variables are found to play very little role in predicting the directional change in the return on the UK stock market index. Our preliminary analyses lead us to choose Model (3) as our benchmark model.

We define Model (5) and Model (6) as follows: Model (5), denoted $X_{5,t}$, adds the first difference of the conditional country risk, $\Delta\beta_t$, to the benchmark model; Model (6), denoted $X_{6,t}$, adds the first difference of the conditional residual risk, $\Delta\sigma_t$, to the benchmark model. We will now refer to Model (5) as the conditional country risk model:

$$X_{5,t} = \left(1, SP_{t-1}, \Delta r^S_{t-1}, \Delta r^L_{t-1}, \tilde{y}_{t-1}, \Delta DV_{t-1}, \Delta\beta_t\right)'$$

while Model (6) will be referred as the conditional residual risk model:

$$X_{6,t} = \left(1, SP_{t-1}, \Delta r^S_{t-1}, \Delta r^L_{t-1}, \tilde{y}_{t-1}, \Delta DV_{t-1}, \Delta\sigma_t\right)'$$

It is worthwhile to note that, although we use the contemporaneous values of the conditional country risk and the conditional residual risk as regressors in the binary logit estimation, β_t and σ_t are both fitted values computed using the information in the conditional variance–covariance matrices which are estimated using information on past returns at $t-1$ and $t-2$. These values are known to the investor at time t.

The binary logit estimation results for Models (5) and (6) are presented in columns (5) and (6) in Table 2 respectively. The results suggest that while the conditional residual risk is a statistically significant predictor, the conditional country risk appears to play very little role in forecasting the sign on the return on the UK stock market. The McFadden R^2 and the Estrella R^2 for Model (6) show an improvement in the goodness-of-fit of the model over the other specifications.

6. Evaluation of forecasts

In this section, we evaluate the forecast performance—both in-sample and out-of-sample—of the benchmark model, the conditional country risk model and the conditional residual risk model. We employ an outcome-based method for evaluating the predictive performance of the model to examine the proportion of correct predictions against the observed outcomes. For each of the three models, we construct a contingency table — a 2×2 matrix — cross-classifying the actual and predicted outcomes in each of the two categories.

The ratio of correct predictions to actual outcomes (also called the 'hit ratio') for the direction of change in the return on the UK stock market is calculated as follows:

$$CP = CP_{\text{Up}} + CP_{\text{Down}} = \frac{1}{T}\sum_{t=1}^{T}\mathbf{1}(\hat{q}_t = q_t) \tag{13}$$

where CP_{Up} is the proportion of correct predictions for the 'upwards' moves, CP_{Down} is the proportion of correct predictions for the 'downwards' moves and $\mathbf{1}(\cdot)$ is the indicator function, taking the value of 1 when the argument is true and 0 otherwise. We show the functional forms of CP_{Up} and CP_{Down} in Appendix A.3. Furthermore, we also employ the Pearson chi-squared test to test for statistical dependence between the predicted and observed outcomes. The null hypothesis of no statistical dependence is tested against the alternative hypothesis. The functional form of the test statistic is presented in Appendix A.4.

6.1. Evaluation of in-sample forecasts

We use the estimated binary logit coefficients to compute the probabilities for each of the two possible outcomes: 'upwards' and 'downwards'. The decision rule described in Eq. (9) is employed to determine the direction-of-change forecast in the return on the UK stock market for each period. The forecasts are then compared against the observed events and a contingency table is constructed for each specification. Although we do not report the contingency tables in our paper, we present the *CP* statistics in Table 3. Our evaluation suggests that all three models perform reasonably well. While the benchmark model correctly predicts the direction of change in the UK stock market 64.9% of the time, the conditional country risk model and the conditional residual risk model performs better with *CP* scores of 66.5% and 67.6% respectively. The Pearson chi-squared test statistics for all three models reject the null hypothesis of no statistical dependence between the observed values and the predicted values, suggesting that these forecasts are not random events and our models are capable of explaining the direction of change in the return on the UK stock market.

To isolate the impact of the conditional country risk and the conditional residual risk from the other variables in the models, we conduct exclusion tests using values from the log-likelihood functions. We compare a model with no restrictions on the coefficient on the variable we wish to test (i.e., the conditional country risk and the conditional residual risk) versus the benchmark model. The likelihood ratio test investigates the null hypothesis that the risk variable is unimportant for prediction of the direction of change in the UK stock market. We also present the results of the exclusion tests in Table 3. The test statistics indicate that, while the coefficient on the conditional country risk can be restricted to zero, the same cannot be said for the coefficient on the conditional residual risk. A zero restriction on the conditional residual risk is therefore rejected. The results suggest that the conditional residual risk is important to the prediction of the UK stock market return sign while the conditional country risk is not.

6.2. Evaluation of out-of-sample forecasts

We evaluate the out-of-sample predictive strength of the models by splitting our sample into two subsamples. The first subsample is used for model estimation while the second subsample is employed for assessing the forecast performance. We use the 'expanding window' method, where the first subsample is expanded—by adding an observation taken from the second subsample—each time a forecast is made. After an observation is added to the first subsample, the model is re-estimated and the process is repeated until the last observation is used.

To mimic the way practitioners use information to make investment decisions, for each out-of-sample forecast we use only the information that is readily available to the investor. For each forecast, the following estimations are performed: Step (i) the output gap is calculated using the Hodrick–Prescott filter; Step (ii) the conditional country risk and the conditional residual risk are computed by fitting the BEKK-GARCH(2, 1) model to the data; and Step (iii) the binary logit models are estimated, using the variables calculated in Step (ii), to obtain the parameters which we then use to make a forecast.

Similar to the in-sample predictions, we evaluate the out-of-sample predictive performance of the benchmark model, the conditional country risk model and the conditional residual risk model. In order to test for robustness, we carry out a total of 10 out-of-sample exercises using different estimation and forecasting windows. We present the out-of-sample results in Table 4. For the first set of out-of-sample forecasts, the first subsample includes the observations from June 1996 to October 2003. The first out-of-sample forecast is made for November 2003, thus allowing us to make a total of 96 out-of-sample forecasts. In each of our subsequent sets of forecasts, the initial estimation window is extended by 6 months and the forecasting window is shortened by 6 months and so on. The last set of exercise, therefore, uses the observations from June 1996 to April 2008 as the initial estimation window to make the first out-of-sample forecast for May 2008.

It is clear that the out-of-sample forecast performance of the conditional residual risk model is far superior to that of the conditional country risk model and the benchmark model. For example, when the forecasting window include observations from November 2005 to October 2011, the benchmark model and the conditional country risk model make 45 and 44 correct predictions, respectively, while the residual risk model accurately predicts

Table 3
The in-sample predictive performance.

	Model		
	(3)	(5)	(6)
CP_{Down}	61	62	60
	(67.0%)	(68.1%)	(65.9%)
CP_{Up}	59	61	65
	(62.8%)	(64.9%)	(69.1%)
$CP_{Overall}$	120	123	125
	(64.9%)	(66.5%)	(67.6%)
Pearson *chi-squared*	16.447**	20.187**	22.797**
d.f.	1	1	1
Prob.	0.000	0.000	0.000
$\log L_{RS}$	n/a	−114.872	−114.872
$\log L_{UN}$	n/a	−114.346	−111.905
Exclusion test *chi-squared*	n/a	1.052	5.934**
d.f.	n/a	1	1
Prob.	n/a	0.305	0.015

Note: Standard errors are shown in parentheses. A total of 185 observations are used in the logit estimations. The sample period starts from 06/1996 to 10/2011. The symbol * indicates statistical significance at the 10% significance level. The symbol ** indicates statistical significance at the 5% significance level.

Table 4
The out-of-sample predictive performance.

Forecasting window (number of forecasts)	Model					
	(3)		(5)		(6)	
	$CP_{Overall}$ (%)	Pearson *chi-squared*	$CP_{Overall}$ (%)	Pearson *chi-squared*	$CP_{Overall}$ (%)	Pearson *chi-squared*
11/2003–10/2011 (96)	59.38	3.35*	57.29	1.98	61.46	4.96**
5/2004–10/2011 (90)	61.11	4.50**	57.78	2.18	63.33	6.37**
11/2004–10/2011 (84)	59.52	3.13*	57.14	1.77	63.10	5.73**
5/2005–10/2011 (78)	60.26	3.34*	60.26	3.34*	65.35	7.39**
11/2005–10/2011 (72)	62.50	4.50*	61.11	3.56*	69.44	10.92**
5/2006–10/2011 (66)	62.12	3.88*	59.09	2.18	68.18	8.80**
11/2006–10/2011 (60)	61.67	3.22**	60.00	2.35	68.33	8.02**
5/2007–10/2011 (54)	62.96	3.60**	62.96	3.60*	66.67	5.93**
11/2007–10/2011 (48)	64.58	4.05**	66.67	5.37**	68.75	6.86**
5/2008–10/2011 (42)	62.90	2.41	64.29	3.50*	66.67	4.84**

Note: Standard errors are shown in parentheses.
The symbol * indicates statistical significance at the 10 significance level.
The symbol ** indicates statistical significance at the 5 significance level.

the direction of change in the UK stock market 50 out of 72 occasions. The ratio of correct predictions against observed outcomes for the conditional residual risk model is approximately 70%. The null hypotheses of independence between the observed and the predicted outcomes are rejected at the conventional significance levels for all three models, suggesting that the correct forecasts made by the models are not random events.

7. Return simulations and trading strategies

In this section we test the usefulness of our models for 'timing' the markets. We compare the return from a passive 'buy and hold' investment strategy to the returns generated from tactical asset allocation strategies guided by the benchmark model, the conditional country risk model and the conditional residual risk model. We follow Leung et al. (2000) and Nyberg (2011) by adopting a simple trading simulation where the investor decides at the beginning of each month whether to invest in the stock market or the money market. Once an investment decision is made, the funds remain invested in that market until the end of the month before a decision is made again at the beginning of the following month. Each investment decision is guided by the forecast of the direction of change in the return on the stock market: if the model forecasts a positive return on the stock market, the investor allocates all of the assets in the stock market; if the model predicts a negative return, the investor then puts the assets into risk-free Treasury bills.[8] It is worthwhile to note that assets are moved between the two markets only when a return sign forecast is different from the forecast made in the previous month.

Following Pesaran and Timmerman (1995) and Nyberg (2011), we take into consideration transaction costs by assuming a charge of ζ_s = 0.50% when assets are shifted from the risk-free rate investment to stocks and a charge ζ_b = 0.10% when assets are shifted from stocks to the risk-free rate investment. The assumptions of transaction costs impose penalties when asset allocation changes between stocks and risk-free rate investments in our simulations. In other words, the simulated portfolio incurs a loss of 0.10% when the allocation is shifted from stocks to risk-free rate investments and a loss of 0.50% when the assets are moved from the risk-free rate investments to stocks.

The decision rule defined in Eq. (9) is also modified to permit the 'degree of confidence' in the estimated probabilities to play a role in determining the directional forecast. When the threshold is assumed to be 0.50, as suggested by Eq. (9), a forecast of the direction of change is made based on a single threshold without taking into account the distance between the threshold and the estimated probability. For instance, estimated probabilities of 0.51 and 0.99 would result in the same directional forecast. This could be undesirable as the estimated probability is sometimes not large (or small) enough to warrant a change in asset allocation. To better utilise the estimated probabilities supplied by the binary logit model, we follow Leung et al. (2000) by adopting what they call 'multiple threshold trading' strategies.

Let γ, where $0.5 \leqslant \gamma \leqslant 1$, be the coefficient of reliability the investor would like to maintain in the trading. The value of γ can be perceived as the extent to which the investor is confident about the predictive strength of the forecasting models or the degree of risk she would like to take. The smaller the value of γ, the less confident the investor is of the models or the more risk averse the investor is. The modified decision rules, incorporating the multiple threshold trading strategies, are:

$$\begin{aligned} &\widehat{P}_0 > (0.5/\gamma) \Rightarrow \text{Short the stock index and long the Treasury bills} \\ &\widehat{P}_1 > (0.5/\gamma) \Rightarrow \text{Long the stock index and short the Treasury bills} \\ &\widehat{P}_0 < (0.5/\gamma) \text{ or } \widehat{P}_1 < (0.5/\gamma) \Rightarrow \text{Do nothing} \end{aligned} \tag{14}$$

We simulate the investment return series by first forecasting the direction of change in the UK stock market out-of-sample using the observations starting from June 1996 up until October 2008. The first forecast of the directional change in return on the UK stock market index is made for November 2008. In each iteration, we compute the probabilities and compare them to the thresholds

[8] Note that since both returns are denominated in US dollars, shifting assets between the two asset classes does not expose the investor to exchange rate risk.

calculated over the range of possible values of γ. Assets are allocated in the appropriate market according to the decision rules in (14) and the investment return is calculated at the end of the period. The estimation window is expanded and the process is repeated again until the last forecast is made. For each specification, we produce a total of 36 simulated return observations.

Our simulations reveal that the holding-period-return on a 'buy and hold' portfolio tracking the UK stock market index during this period equals 33.33%. The annualised return, compounded geometrically, equals 10.06% per year. During the same period, the annualised investment returns achieved by pursuing three different trading strategies, guided by our econometric models, range from approximately 5.50% to 13.32% as shown in Table 5. The exact values depend on the values of γ used in the calculation of the thresholds. Nonetheless, our simulation results suggest that, when $0.50 \leqslant \gamma \leqslant 0.75$, the conditional residual risk model is superior to the conditional country risk model and the benchmark model. The results presented in Table 5 suggest that, by setting the value of γ equal to 0.85 for the benchmark model and 0.75 for both the conditional country risk model and the conditional residual risk model, the maximum returns on investments achieved by following the trading recommendations are 12.60%, 11.43% and 13.32% respectively. The value of γ which maximises the return in our simulations is also found by Leung et al. (2000) to be around 0.70. Even though they caution that the range over which γ should be chosen is an empirical issue, they recommend a trader to choose a γ at around 0.70.

The level of return is not the only target money managers strive to meet, but they are also often judged by the amount of risk their managed portfolios are exposed to. One of the many investment objectives pursued by asset managers is to achieve a level of risk-adjusted return, often measured by the Sharpe ratio, specified by their clients' risk and return appetites. The ratio, originally proposed by Sharpe (1966) and subsequently revised by Sharpe (1994), measures the excess return (or risk premium) per unit of deviation in the return on investment. It has the following functional form:

$$SR = \frac{\overline{R} - \overline{R}_f}{\hat{\sigma}} \tag{15}$$

where $\overline{R}$ is the average portfolio return, $\overline{R}_f$ is the average return on the risk-free rate investment and $\hat{\sigma}$ is the standard deviation of the portfolio return.

The Sharpe ratios for each investment strategy are presented in Table 6. Except for the passive buy and hold portfolio, the ratio for each trading strategy is calculated based on the value of γ that maximises the portfolio return: we set $\gamma = 0.85$ for the benchmark model and $\gamma = 0.75$ for both the conditional country risk model and the conditional residual risk model. According to Table 6, it is evident that the conditional residual risk model generates the highest risk-adjusted return compared to the Sharpe ratios for the other three investment strategies. It should also be noted that, compared to the actively managed portfolios guided by the three investment strategies, the buy and hold portfolio carries significantly higher risk. This should come as no surprise as the markets have been highly volatile since the collapse of Lehman Brothers in September 2008.

To conclude this section, we test the hypotheses that differences in the Sharpe ratios—calculated on the performance of our simulated trading strategies—are statistically significant. We employ two test statistics: (i) the test statistic of Jobson and Korkie (1981) which has been corrected by Memmel (2003) and (ii) the test statistic of Ledoit and Wolf (2008) that uses heteroscedasticity and autocorrelation consistent (HAC) standard errors. The former test statistic is constructed based on an assumption that the return distribution is normally distributed; the latter test statistic does not assume normality and is therefore more robust to serial correlation. We compute both test statistics to test whether the risk-adjusted return generated from the trading strategy guided by the conditional residual risk model is significantly different from the risk-adjusted returns generated from the other three trading strategies.[9]

Table 5
The simulated annualised returns based on multiple threshold trading strategies.

γ	Model		
	(3)	(5)	(6)
0.50	10.06	10.06	10.06
0.55	8.97	8.97	10.71
0.60	8.97	8.97	13.21
0.65	8.97	8.97	13.21
0.70	8.97	8.97	13.21
0.75	11.43	11.43*	13.32*
0.80	11.43	11.43*	8.41
0.85	12.60*	6.60	9.55
0.90	11.54	9.86	9.55
0.95	5.45	6.99	6.54
1.00	7.01	7.97	6.54
Average	9.74	9.30	10.34

Note: The symbol * indicates the maximum value.

Table 6
The Sharpe ratios.

	Model			
	Buy and hold	(3)	(5)	(6)
Y	N/A	0.85	0.75	0.75
$\overline{R}$	0.5747	0.6329	0.5444	0.6863
σ_R	6.6339	4.2666	4.2427	4.2661
Sharpe ratio	0.0849	0.1457	0.1256	0.1582

Note: The arithmetic average monthly return on the risk-free rate is 0.0114%.

The one-sided p-values for the Memmel (2003)'s test statistics computed for the differences between the Sharpe ratio of the conditional residual risk model and the Sharpe ratios of the buy and hold portfolio, the benchmark model, and the conditional country risk model equal 0.01, 0.16, and 0.001, respectively. These results indicate that the risk-adjusted return generated by following trading recommendations of the conditional residual risk model is significantly higher than the risk-adjusted returns generated by the buy and hold portfolio and the conditional country risk model, but not the benchmark model—albeit the fact that the p-value of 0.16 is close to the 0.10 rejection threshold for the 10% significance level.

The test statistics of Ledoit and Wolf (2008) point to a more cautious interpretation of our trading simulation results. The one-sided p-values calculated using the method of Ledoit and Wolf (2008) for the differences between the Sharpe ratio of the conditional residual risk model and the Sharpe ratios of the buy and hold portfolio, the benchmark model, and the conditional country risk model equal 0.29, 0.37, and 0.16, respectively. Although these findings are less than encouraging, they are not unexpected. Since HAC standard errors are robust to the presence of both heteroscedasticity and autocorrelation, they are larger than normal standard errors—leading to smaller test statistics. In fact, it is worth noting that Ledoit and Wolf (2008) also observe significant increases in p-values when testing for differences in the Sharpe ratios of both mutual funds and hedge funds; in their empirical applications,

[9] We thank Michael Wolf for making the R code for the test statistic of Ledoit and Wolf (2008) publicly available. The code can be downloaded from: http://www.econ.uzh.ch/faculty/wolf/publications.html#9.

the p-value increase from 0.01 when using the test of Memmel (2003) on hedge fund data to 0.15 when the HAC standard error is calculated thereby failing to reject the null hypothesis that the difference in the Sharpe ratios is not statistically significant.

8. Conclusions

This paper offers evidence of the usefulness of return volatility in forecasting the direction of change in the asset return. Both in-sample and out-of-sample forecast evaluation indicates that conditional volatility possess statistically significant predictive power and can be employed to forecast the direction of change in asset returns. Our findings are also in line with previously published research that provides support for the role residual risk plays in asset pricing, such as Glosten et al. (1993), Gonzalez-Rivera (1996), Goyal and Santa-Clara (2003), Guo and Savickas (2003), Ghysels et al. (2004), Malkiel and Xu (2004) and Fu (2009). They also lend support to previous studies carried out by Bekaert and Harvey (1995), Carrieri et al. (2007) and Guesmi and Nguyen (2011) that individual stock markets and the global stock market are segmented as the impact of conditional residual risk on asset returns dominates the impact of conditional country risk.

Our methodology involves a two-stage process starting with the estimation of the conditional volatility that takes into account co-movement between returns on the country stock market and returns on the global stock market. This approach yields improvements over other univariate and multivariate approaches that assume constant correlation since market interdependence tends to rise during periods of high uncertainty. Using data on the UK stock market index and the world market index between 1996 and 2011, the conditional volatility is modelled using the BEKK-GARCH approach. This technique ensures that the variance–covariance matrices are positive definite, thus guaranteeing convergence when the model is estimated by maximum likelihood. Using the information obtained from the estimated conditional variance–covariance matrices, we compute the conditional country risk and the conditional residual risk, which subsequently enter the binary logit estimation in the second stage as explanatory variables.

In the second stage, we employ the binary logit model to investigate the ability of the conditional country risk and the conditional residual risk to predict the direction of change in the UK stock market index. The binary logit estimation results indicate that the parameters of the output gap, the dividend yield and the conditional residual risk are all statistically significant. The outcome-based in-sample forecast evaluation suggests that, while the conditional country risk possesses little predictive strength, the conditional residual risk appears to be a competent predictor. It correctly forecasts the sign of the return on the UK stock market 125 out of 185 occasions compared to 120 and 123 occasions for the benchmark model and the conditional country risk model. The results from the exclusion tests, which seek to examine the marginal predictive strength of the variable that is excluded from the model, show that the conditional country risk plays very little role in forecasting the direction of change in the return on the UK stock market whereas the conditional residual risk is shown to be an important explanatory variable and should not be excluded from the forecasting model. The out-of-sample forecast evaluation also suggests that the conditional country risk model dominates the other models when used to forecast the sign of return on the UK stock market out-of-sample.

Finally, we conduct trading simulations which use the benchmark model, the conditional country risk model and the conditional residual risk model to produce trading recommendations. Our simulations are enhanced by the use of multiple thresholds and incorporate transaction costs that impose a penalty on the portfolio return when assets are shifted between the risky and the risk-free investments. Our simulation results indicate that an actively managed portfolio which follows a trading strategy guided by the conditional residual risk model outperforms portfolios guided by the other trading strategies. Moreover, the portfolio managed using the conditional residual risk model generates the highest Sharpe ratio, suggesting that the trading strategy succeeds in enhancing the portfolio return while reducing the overall level or risk.

Appendix A

A.1

In the BEKK-GARCH model the parameters of the conditional variance and conditional covariance equations are estimated simultaneously by maximising the following log-likelihood function:

$$\ln L(\theta) = -\frac{1}{2}\sum_{t=1}^{T}\left(\ln(|\mathbf{H}_t|) + \boldsymbol{\varepsilon}_t'\mathbf{H}_t^{-1}\boldsymbol{\varepsilon}_t\right) \tag{16}$$

where $\mathbf{H}_t$ is the conditional covariance matrix and $\boldsymbol{\varepsilon}_t$ is the GARCH error vector at time t.

A.2

The parameters in the binary logit model can be estimated by maximising the following log-likelihood function:

$$\ln L(\beta) = \sum_{t=1}^{T}\sum_{j=0}^{1}\mathbf{1}[q_j = j]\Pr(q_t = j|X_t) \tag{17}$$

where $\mathbf{1}[\cdot]$ is the indicator function.

A.3

The proportions of correct predictions for the 'upwards' moves and the 'downwards' moves are computed as follows:

$$CP_{\text{Up}} = \frac{\sum_{t=1}^{T}\mathbf{1}(\hat{q}_t = 1)\mathbf{1}(q_t = 1)}{\sum_{t=1}^{T}\mathbf{1}(q_t = 1)}$$

and

$$CP_{\text{Down}} = \frac{\sum_{t=1}^{T}\mathbf{1}(\hat{q}_t = 0)\mathbf{1}(q_t = 0)}{\sum_{t=1}^{T}\mathbf{1}(q_t = 0)}$$

where q_t is an actual outcome and $\hat{q}_t$ is a forecasted outcome.

A.4

The null hypothesis for the test of association in a 2×2 contingency table is: H_0: No association between the predicted and the observed values. We reject H_0 if

$$\sum_{i=1}^{2}\sum_{j=1}^{2}\frac{(O_{ij} - \widehat{E}_{ij})^2}{\widehat{E}_{ij}} > \chi_1^2$$

where O_{ij} is the number of observations in the cell that is in the ith row and jth column and where

$$\widehat{E}_{ij} = \frac{R_i C_j}{n}$$

where R_i and C_j are the corresponding row and column totals.

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